

Linear algebra – 7

Euclidean spaces, orthogonality and the SVD

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Theoretical minimum.

Def. An operator A on a Euclidean (unitary) space is *self-adjoint* if $A^* = A$, *orthogonal (unitary)* if $A^*A = E$, and *normal* if $A^*A = AA^*$. Here $A^* = \overline{A}^T$.

Theorem 1 (spectral theorem). A complex matrix is normal if and only if it is unitarily diagonalisable; a real symmetric matrix is orthogonally diagonalisable with real eigenvalues. Consequently a real orthogonal matrix is orthogonally similar to a block-diagonal matrix with blocks ± 1 and plane rotations.

Theorem 2 (singular value decomposition). Every $A \in M_{m,n}(\mathbb{R})$ can be written as $A = U\Sigma V^T$ with U, V orthogonal and $\Sigma = \text{diag}(\sigma_1 \geq \dots \geq \sigma_r > 0)$ padded by zeros; the σ_i are the square roots of the nonzero eigenvalues of $A^T A$. Geometrically A maps the unit sphere onto an ellipsoid with semiaxes σ_i .

Theorem 3 (polar decomposition). Every $A \in M_n(\mathbb{R})$ factors as $A = QS$ with Q orthogonal and $S = \sqrt{A^T A}$ positive semidefinite; S is unique, and Q is unique when A is invertible.

Fact 1. $\sigma_1 = \|A\| = \max_{\|x\|=1} \|Ax\|$ and $\sigma_k = \min_{\dim L=n-k+1} \max_{x \in L, \|x\|=1} \|Ax\|$ (the minimax principle).

1. Prove the spectral theorem for a real symmetric operator, deduce the variational description $\lambda_{\max} = \max_{\|x\|=1} \langle Ax, x \rangle$, and show that a normal matrix with real eigenvalues is symmetric.
2. (SEEMOUS 2021) Let $A \in M_n(\mathbb{C})$ satisfy $(AA^*)^2 = A^*A$.

- a) Prove that $AA^* = A^*A$.
 - b) Prove that every nonzero eigenvalue of A has modulus 1.
3. (IMC) Let A be a real 3×3 matrix such that the vectors Au and u are orthogonal for every column vector $u \in \mathbb{R}^3$. Prove that $A^T = -A$, and that there is $v \in \mathbb{R}^3$ with $Au = v \times u$ for all u .
 4. (ФКН, веча 2021) A real $n \times n$ matrix is given whose first k rows are unit vectors, pairwise orthogonal, and whose first l columns are also unit vectors, pairwise orthogonal. Prove that the remaining entries can be changed so that the matrix becomes orthogonal.
 5. (IMC 2025 shortlist) Let $P, Q, R \in O(3)$ be real orthogonal 3×3 matrices, i.e. $P^T P = Q^T Q = R^T R = E_3$. Show that the equation $P + Q = R$ has no solutions in $O(3)$.